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the Compressor discharge through the Natural Gas By-Pass Cooler (E-308).

2.5 SYNTHESIS

This section catalytically combines hydrogen and nitrogen to produce ammonia and separates the ammonia from the remaining gas stream.

Synthesis Gas containing hydrogen, nitrogen, ammonia and inerts, that are methane and argon, is passed over the catalyst in the Ammonia Converter (R-401) where a part of the hydrogen and nitrogen react to form more ammonia. The effluent from the Ammonia Converter is cooled in a series of heat exchangers to condense product ammonia. The remaining gas, after separation of the liquid ammonia, is boosted up by the recycle stage. The recirculated synthesis gas is cooled in a series of heat exchangers, including two stages of the Ammonia Cooled Condenser, and make-up gas is introduced into and mixed with the recycle gas in the Secondary Ammonia Cooled Condenser. Another portion of ammonia is condensed in this series of heat exchangers, and after separation of the liquid ammonia, the synthesis loop is completed with the gases being again introduced into the Ammonia Converter.

a. Ammonia Conversion

The Ammonia Converter consists of an inner shell (the catalyst basket) containing three separate catalyst beds, and an outer pressure-containing shell with a resultant annular space between the two shells. The major part of the synthesis gas from the Feed Preheater (E-403) is passed through the annular space in order to maintain a low temperature on the outer shell. The remaining gas is available as quench gas to control the catalyst temperature in the second and third beds of the Converter.

The gas leaving the Converter annular space is heated to about 400°C in the shell side of the Converter Exchanger (E-401) before being introduced into the first catalyst bed of the Converter. Temperatures in the first bed are controlled with by-passing of the Feed Preheater, the Converter annular space, and the Converter

Exchanger. While passing over the catalyst in the first bed a part of the hydrogen and nitrogen react to form ammonia with the evolution of heat and a resultant temperature rise. The hot effluent from the first catalyst bed is quenched (i.e. cooled) by mixing with a part of the cooler gas from the Feed Preheater before the mixture passes through the second catalyst bed. In a similar manner the effluent from the second bed is quenched before passing to the third bed. The converted gas leaving the Converter is cooled in the Converter Exchanger, and

Exchanger. While passing over the catalyst in the first bed a part of the hydrogen and nitrogen react to form ammonia with the evolution of heat and a resultant temperature rise. The hot effluent from the first catalyst bed is quenched (i.e. cooled) by mixing with a part of the cooler gas from the Feed Preheater before the mixture passes through the second catalyst bed. In a similar manner the effluent from the second bed is quenched before passing to the third bed. The converted gas leaving the Converter is cooled in the Converter Exchanger, and further cooled in the Boiler Feedwater Preheater (E-402) before entering the tube side of the Feed Preheater (E-403) where it heats the synthesis gas going to the Converter.

b. Primary Condensing (Water Cooled)

The converted gas leaving the Feed Preheater flows through the Water Cooled Condensers (E-404 A,B) where about 35% of the Ammonia product is condensed. The condensed liquid ammonia is then separated from the recycle gas in the Primary Separator (V-401). The liquid ammonia is let down, decreasing to about 21 Kg/cm^2G, as it flows into the Letdown Tank (V-409). The remaining gas leaving the Primary Separator returns to the Compression Section as recycle gas.

A part of the recycle gas is extracted as the purge gas to maintain inert gas level in the recycle gas at about 10%. To recover ammonia vapor contained in the purge stream, the gas is chilled in the Purge Condenser (E-408) by liquid ammonia from the Refrigeration Receiver (V-408) in the Refrigeration Section. The condensed ammonia is separated in the Purge Gas Separator (V-403) and flows into the Letdown Tank. The purge gas from the Purge Gas Separator is warmed in the Purge Gas Jacket Heater (E-416) flowing to the High Press. Boiler Unit.

c. Secondary Condensing (Ammonia Cooled)

Recycle gas from the circulator section of the Synthesis Gas Compressor is cooled to 26.5°C in the shell side of the Cold Exchanger (E-407) by heat exchange with the cold gas from the Secondary Separator (V-402). The cooled recycle stream, containing a small quantity of condensed ammonia, is further cooled to 14°C in the Primary Ammonia-Cooled Condenser (E-405), condensing more ammonia. The cooled stream from the Primary Ammonia-Cooled Condenser is mixed with make-up gas from the Synthesis Gas Compressor, in the Secondary Ammonia Cooled Condenser (E-406). The mixture is then cooled to 2.2°C, and more ammonia is condensed so that approximately 4% ammonia remain in the gas phase. Approximately 65% of the product ammonia is condensed in the Cold Exchanger and the two Ammonia Cooled Condensers. E-407 E-405 & 406

Liquid ammonia is separated from the gas stream in the Secondary Separator (V-402) and then sent to Letdown Tank (V-409). The mixed synthesis gas vapor (at about 2.2°C) from the Secondary Separator returns to the tube side of the Cold Exchanger for reheating to 50°C. The synthesis gas then passes through the shell side of the Feed Preheater for further heating to 148°C before entering the Ammonia Converter (R-401).

Product ammonia collected in both the Primary and Secondary Separators is let down in pressure to 21 Kg/cm^2G in the Letdown Tank (V-409) where the dissolved gases are flashed from the product before it is sent to the Refrigeration Section.

The flashed inert gases are cooled in the Ammonia Vent Condenser (E-409) to condense ammonia flashed over with the gases, and liquid ammonia is separated in the Vent Separator (V-404). Liquid ammonia from Vent Separator flows to the First Stage Suction Separator (V-405). The separated gas is combined with the purge gas to be sent to the H.P. Boiler Unit.

Liquid ammonia from the Letdown Tank is introduced through the Ammonia Cooled Condensers into the Refrigeration Section. Product ammonia of -32°C is pumped to the Ammonia Storage Tank from the 1st Stage Suction Separator (V-405) by the Ammonia Product Pump (P-401AB).

2.6 REFRIGERATION

a. Process

The Refrigeratiqn Section provides liquid ammonia refrigerant to cool synthesis gas process stream to lower temperatures by means of liquid ammonia evaporation at three pressure levels:

- 4.9 Kg/cm^2G Level

  In the Primary Ammonia Cooled Condenser (E-405) refrigerant liquid ammonia from the Receiver (V-408) and the product ammonia from the Letdown Tank (V-409) are partially vaporized at about 8.0°C and 4.9 Kg/cm^2G.

- 2.8 Kg/cm^2G Level

  In the Secondary Ammonia Cooled Condenser (E-406) liquid ammonia cascaded from the Primary Ammonia Cooled Condenser is partially vaporized at about -4.0°C and 2.9 Kg/cm^2G.

- 0.2 Kg/cm^2G Level

  In the Purge Condenser (E-408) and in the Ammonia Vent Condenser (E-409) liquid ammonia from the Refrigeration Receiver is vaporized at about -30.0°C and 0.2 Kg/cm^2G.

Vaporized ammonia, according to the pressure level, is passed through one of three stages of the Suction Separators (V-405, 406 and 407) to separate liquid before entering the Refrigeration Compressor (K-401). Ammonia liquid-vapor comes from any of the following places:

E-405

1) Ammonia vapor from the Primary Ammonia Cooled Condenser passes through the Third Stage Suction Separator (V-407) before entering at an side-stream suction point on the high pressure case of the Refrigeration Compressor. The Third Stage Suction Separator normally operates without liquid flow; if a liquid level does develop, it is cascaded into the Second Stage Suction Separator.

E-406

2) Ammonia vapor from the Secondary Ammonia Cooled Condenser passes through the Second Stage Suction Separator (V-406), where it combines with the discharge gas flow from the Refrigeration Compressor low pressure case, before entering the first suction point on the high pressure case of the Refrigeration Compressor. The Second Stage Suction Separator normally operates without liquid flow, but if a liquid level develops, it cascades into the First Stage Suction Separator.

E-408 E-409

3) Ammonia vapor from the Purge and Ammonia Vent Condensers, combined with the flashed vapors from various liquid streams, passes through the First Stage Suction Separator (V-405) before entering the suction of the low pressure case of the Refrigeration Compressor. Ammonia liquid at -4.0°C cascaded from the Secondary Ammonia Cooled Condenser, condensed ammonia from the Vent Separator (V-404) and the Purge Gas Separator (V-403) are normally flashed into the First Stage Suction Separator. Liquid not flashed is pumped from the First Stage Suction Separator, by the Ammonia Product Pump (P-401 A or B), to the Ammonia Storage Tank as product.

b. Compression

The Refrigeration Compressor is centrifugal type driven by steam turbine and consists of two casings.

The pressure in each of the three levels of ammonia evaporation is maintained by the Refrigeration Compressor (K-401). The Compressor draws its feed from the three suction separators which feed the low and high pressure cases of the machine. The low pressure case draws vapor from the First Stage Suction Separator (V-405) at 0.07 Kg/cm^2G and discharges the vapor into the Second Stage Suction Separator (V-406). The high pressure case, with suction at two pressure levels draws vapors from the Second Stage Suction Separator (V-406) at 2.5 Kg/cm^2G and from the Third Stage Suction Separator (V-407) at 4.4 Kg/cm^2G. The final discharge vapor from the Compressor is cooled and condensed to liquid ammonia in the Ammonia Condenser (E-412). The liquid ammonia flows by gravity into the Refrigeration Receiver (V-408) from where it commences the evaporation cycle again by going to the Ammonia Cooled Condenser (E-405).
The first suction pressure is controlled by the speed of turbine. To ensure that each stage of the Refrigeration Compressor has sufficient vapor flow to prevent surging of the machine anti-surge control valves are provided on both cases to recycle compressed ammonia vapor and desuperheating with liquid ammonia from the Refrigeration Receiver (V-408).

The flow required to avoid surge in the first stage stream, as measured at the first stage discharge, is maintained by the introduction of desuperheated gas into the First Stage Suction Separator. The flow required to avoid surge in the second stage stream, measured at the second stage suction, is maintained by introducing desuperheated gas into the Second Stage Suction Separator.

The flow required to prevent surge in the third stage stream, measured at the third stage discharge, is maintained by introducing desuperheated gas into the Third Stage Suction Separator.